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Breast cancer: treatment & follow-up care

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Treatment

For several decades, surgery has not been the only option. There are now several possibilities for adjuvant and neoadjuvant treatments, as well as effective palliative care.

Nowadays, the challenge lies in skillfully combining and scaling several treatment types, including several drug groups. This has led to changes in local surgical treatment strategies, with surgical procedures now being one building block of a global treatment concept.

> Modern treatment planning for breast cancer is carefully adapted to the patient and the individual characteristics of the tumor, with general treatment regimens providing guidelines (of limited use individually).

Individual treatment decisions are based on the general clinical situation and the patient's general status, as well as a wide range of clinical, morphological, immunohistochemical and molecular genetic tumor characteristics.

The following form an indispensable foundation for the modern (usually multimodal) approach:

- Menopause status; age on first diagnosis
- Axillary lymph node status (TNM)
- Tumor size (TNM)
- Histological type (WHO), grading (G), lymphatic/venous invasion, proliferation markers
- Presence of distant metastases (TNM)
- Hormone receptor status ER, PR; HER2 status
- If applicable other molecular prognostic/predictive markers

These tumor characteristics, taken together as a whole, define the prognostic subgroup to which the patient belongs.

Predictive factors include:

- ER/PR status
- HER2 status
- complete pathological response (pCR) following neoadjuvant chemotherapy

For the individual subgroups, international specialists meet regularly at consensus conferences to discuss the published data and further refine treatment guidelines.

Four starting points for categorizing the treatment of breast cancer

Operable (early) breast cancer

Synonym: previously referred to as primary breast cancer.

Breast cancer often presents at a stage which is operable, with no identifiable distant metastases. Initial treatment is surgery, if necessary following preoperative chemotherapy or antihormone tumor shrinkage. This must be followed by adjuvant treatment (chemotherapy and/or hormone therapy and/or immunotherapy and/or radiation).

Primary inoperable, locally advanced cancer

Initial diagnoses in locally advanced stages are rarer. Neoadjuvant chemotherapy or neoadjuvant antihormone therapy are often used for tumor shrinkage. Alternatively, a mastectomy (ablation) can be discussed with the patient. Other treatment options then follow.

Metastasized breast cancer

In general, there is no curative treatment. Instead, palliative care is offered. Maintaining quality of life is the main objective. If possible, prolongation of life may also be a goal. To select an efficient form of treatment that nevertheless entails the fewest possible side effects, knowledge of the detailed biological tumor characteristics is required (hormone receptor status (ER, PR), HER2 status, Ki-67 if necessary, further markers).

Locoregional recurrence ("local recurrence")

In case of local recurrence (breast, thoracic wall, axilla) and/or locoregional metastasis (regional lymph nodes), all local treatment options (re-operation, radiation) are first exhausted and systemic therapy (antihormone therapy, immunotherapy, chemotherapy) may be indicated.

Surgical procedures

Treatment for breast cancer is multimodal. The different specialties involved work together.

The surgical strategy for breast cancer has changed fundamentally in recent decades. While modified radical mastectomy was preferred as the safe standard procedure until 1980, today, the specificities of individual cases are used to make an individual treatment decision. Today, organ loss can often be avoided, without having to make compromises in terms of survival.

Prospective randomized studies (with an observation period of > 20 years!) show the non-superiority of mastectomy versus breast-conserving treatment with regard to survival.

Modern surgical procedures, in combination with adjuvant treatment, achieve local disease free survival with minimal loss of physical integrity.

All surgical procedures are supplemented with adjuvant treatments, based on the individual risk constellation.

Adjuvant treatment refers to any treatment after surgical excision of the primary tumor where there is no distant metastasis. This is prophylactic treatment to reduce the risk of recurrence.

Table 1: Overview of possible forms of adjuvant treatment (more detail is given in the following sections)

Adjuvant radiotherapy (RT)	RT of remaining breast tissue RT of lymph drainage pathways for affected lymph nodes RT of thoracic wall (only after mastectomy at high risk of thoracic wall recurrence)
Adjuvant chemotherapy	Polychemotherapy

Adjuvant hormone therapy	Selective estrogen receptor modulators (SERM) Aromatase inhibitors in postmenopause GnRH analogues plus aromatase inhibitors in premenopausal woman with high risk of recurrence
Adjuvant immunotherapy	For HER2+ tumors, the monoclonal antibodies trastuzumab/ pertuzumab or T-DM1 reduce the rate of recurrence.
Adjuvant CDK4/6 inhibitors	In HR+/HER2 breast cancer and aggressive tumours with a high risk of recurrence, the addition of CDK 4/6 in combination with aromatase inhibitors or AI and GnRH reduces the recurrence rate

Recently, CDK 4/6 inhibitors (i.e. the agent blocks cyclin-dependent kinases 4 and 6 (CDK 4/6), which normally regulate cell division and cell growth) have also been used in the adjuvant situation (see palliative medical therapy).

Polychemotherapy: for moderate to high risk: anthracycline plus taxane/carboplatin-based [P35]

Surgical procedures involving the breast

Breast-conserving surgery (BCS)

Today, breast-conserving surgery is practiced where possible. The feasibility of a breast-conserving procedure depends on the size of the breast versus the tissue area to be resected: A result without significant breast deformation can only be expected if there is sufficient remaining breast tissue. Only then can this approach be offered.

Overview:

1. Initially in specific cases reduction of tumor size before surgery: down-staging with neoadjuvant systemic treatment, typically chemotherapy and/or antihormone therapy. This can lead to a more favorable tumor size/breast volume ratio before surgery.
2. Lumpectomy/segmentectomy/quadrantectomy/oncoplastic reduction or tumorectomy (“breast-conserving surgery”) ...
3. ... including axillary lymph node staging (see below “Surgical procedures involving the axilla”)
4. Then mandatory: Adjuvant radiotherapy of remaining breast tissue and

depending on level of lymph node involvement lymph drainage areas if necessary.
5. In some cases: Adjuvant systemic treatment (hormone therapy, immunotherapy, chemotherapy).

Mastectomy

There are various forms of mastectomy:

- Ablation (breast tissue including skin and nipple removed)
- Skin-sparing mastectomy (skin left intact, removal of nipple and breast tissue)
- Nipple-sparing mastectomy (skin including nipple left intact, removal of breast tissue only)
- indicated for pronounced or multicentric large tumors
- indicated for small breasts (for technical surgical or cosmetic reasons)
- indicated on basis of patient desire
- indicated for inflammatory breast cancer

Overview:

1. Mastectomy: A mastectomy removes the pectoral fascia and overlying breast tissue, potentially including the skin/nipple depending on the technique. The pectoralis muscle is left intact.
2. ... including axillary lymph node staging (see below “Surgical procedures involving the axilla”)
3. Subsequently in high-risk situations: Adjuvant thoracic wall radiotherapy.
4. In some cases: Adjuvant systemic treatment (hormone therapy, immunotherapy, chemotherapy).

High-risk situations: i.e. tumor diameter > 5 cm, or resection margin involvement, or axillary lymph node involvement (generally ≥ 1–3 involved lymph nodes)

Surgical procedures involving the axilla

Axillary surgery has undergone a paradigm shift in breast-conserving surgery in recent years. Previously, an axillary dissection was carried out for histologically confirmed lymph node involvement.

> The histological clarification and clip marking of possible axillary lymph node involvement is inevitable before further treatment is planned: neoadjuvant treatment, surgery.

Two procedures may be used depending on the number of axillary lymph nodes involved:

Complete axillary lymphadenectomy = axillary dissection

english: axillary lymph node dissection, ALND

Lymph node removal is the removal of Level II and II (if necessary Level III) lymph nodes (at least 10). If possible, a core biopsy should be taken for histological testing before any further decisions are made on surgical treatment in case of clinically suspicious lymph nodes. ALND is the standard procedure for breast-conserving surgery from involvement of three or more SLN, and from involvement of one SLN for simultaneous mastectomy.

There are currently several international studies, including the TAXIS trial, investigating whether tailored axillary dissection (removal of sentinel lymph nodes, palpable enlarged lymph nodes, initial positive confirmed lymph nodes) is sufficient with subsequent radiotherapy to avoid complete axillary dissection in case of lymph node involvement. The Alliance Trial A011202 also includes patients with residual tumor in the sentinel lymph nodes after neoadjuvant chemotherapy. Patients were randomized into a control group with axillary dissection and an experimental group with axillary radiotherapy. All received extensive supraclavicular and parasternal radiotherapy. This study will clarify whether residual tumor can be left in the axilla following systemic pre-treatment, with subsequent regional radiotherapy.

However, axillary lymphadenectomy is known for its intrusive consequences and side effects:

- Mild, intermittent lymphedema of the arm (20 to 30 %)
- Severe, disabling lymphedema (roughly 3 %)
- Impaired mobility of shoulder and arms
- Areas of skin on back of upper arm/axilla with hyposensitivity and/or pain syndrome

Sentinel lymph node biopsy

The selective, step-by-step approach is based on the following hypotheses:

- With small breast tumors, lymph drainage is disciplined: it first involves just one group of lymph nodes close to the tumor location. This first group of lymph nodes in the direction of drainage are called the sentinel lymph nodes.

- If the sentinel lymph nodes are “tumor-free” on biopsy, no involvement of other axillary lymph nodes (further down the drainage pathway) or distant metastasis should be expected. They therefore do not need dissection, thus removing the associated risk of morbidity.

On comparing the two methods, it should be noted that over 50 % of all patients are node-negative and gain no benefit from “routine” axillary lymphadenectomy (see above “Complete axillary lymphadenectomy”) with the aforementioned postoperative morbidity.

The advantages of sentinel lymph node biopsy are:

- less postoperative morbidity
- less surgery time
- no axillary wound drainage
- less postoperative hospitalization

Today, sentinel lymph node biopsy is standard treatment, although a correct indication is essential. The typical indication for sentinel lymph node biopsy is unifocal breast cancer, tumor diameter < 5 cm, with a clinically node-negative patient.

Implementation

Before surgery, Technetium99 labeled human serum albumin is administered as a peritumoral injection. A subsequent optional lymphoscintigraphy detects the individual flow of the (axillary or parasternal) lymph drainage and the first lymph nodes involved. These are identified during surgery using a gamma detector. In addition, a lipophilic blue dye (e. g. patent blue) can be administered as an intraoperative peritumoral injection (combined detection method) to optimize accuracy. In principle, sentinel lymph nodes are all radioactive and/or blue-dyed lymph nodes identified during surgery as having a radioactivity count of at least 10 % of the hottest node.

These SLN then undergo histological testing:

- All excised sentinel lymph nodes are meticulously tested (several sections) again after surgery for macrometastases (> 2mm), micrometastases (0,2–2 mm) and single tumor cells.
- Subsequent findings of metastasis in sentinel lymph nodes lead to further surgery --> now “real” axillary dissection (for > 2 affected SLN and BCS or > 0

affected SLN for mastectomy).

Two studies: IBCSG 23-01 and ACOSOG Z0011 could show that for breast-conserving surgery in a long-term follow-up of 10 years (multicentric phase III non-inferiority study), no axillary dissection was required in clinically node-negative patients with limited tumorous sentinel lymph nodes (normal axilla on palpation, up to two histologically affected lymph nodes, tumor < 5 cm). In these studies, patients with clinically node-negative axilla and tumors < 5cm were randomized into two groups: axillary dissection and an experimental group without axillary treatment (but with the following systemic treatment: chemotherapy, antihormone therapy, breast radiotherapy). The axillary recurrence rate for both groups was under 2%, there was no statistically significant difference with regard to disease-free survival or overall survival [P06, P07].

With max. 2 positive lymph nodes and a tumor size smaller than 5cm, sentinel lymph node biopsy with subsequent radiation and, as applicable, chemotherapy/ antihormone therapy is sufficient. [P06][P07]

Reconstructive plastic surgery procedures

Breast reconstruction following mastectomy can take place as an integrated part of the same surgery (oncoplastic surgery) or as a separate surgery (delayed reconstruction). Both procedures have individual advantages and disadvantages - both psychological and technical (see below “Further information on surgical techniques for breast reconstruction”):

- Often, oncological certainty means oncoplastic surgery is possible. In this case, primary breast reconstruction is carried out immediately after the procedure to resect a tumor (including for recurrent tumors). Immediate reconstruction can be carried out using autologous tissue or implants, with patient satisfaction appearing higher after autologous reconstruction, particularly after radiotherapy. Evidence shows that the morbidity rate of primary breast reconstruction is no higher than that of delayed reconstruction. The restoration of physical integrity and aesthetic appearance can be achieved in a single procedure, as can local tumor control [P02].
- Alternatively, the specific requirements of oncological certainty (if radiotherapy is anticipated), local tissue behavior, patient preference, or medical recommendation may mean a delayed breast reconstruction is necessary. In this case a second procedure is carried out (secondary breast reconstruction following completed primary treatment).

[P02]

Medication

Medication is used in various ways in cases of breast cancer:

- adjuvant treatment - to reduce the risk of recurrence or metastasis
- treatment of existing metastases

Medication takes effect systemically, affecting the whole body. This means it can influence both subclinical ("unidentified") and manifest metastases.

> The various medications used can include chemotherapeutics, various drug groups for hormone therapy (endocrine therapy) and under certain conditions monoclonal antibodies (immunotherapy).

Table 2: Chemotherapy: the most common (most active) cytostatic agents

Anthracyclines	Doxorubicin, Epirubicin
Taxanes	Paclitaxel, Docetaxel, nab-Paclitaxel
Alkylating antineoplastic agents	Cyclophosphamide
Antimetabolites	5-Fluorouracil, Methotrexate, Capecitabine, Gemcitabine
Vinca alkaloids	Vinorelbine

Anthracyclines and taxane are clearly the most important cytostatic agents for breast cancer. They achieve the greatest benefit in polychemotherapy regimens. Note: Anthracyclines and taxanes generally lead to alopecia (--> timely information to patient regarding headscarves/wigs).

Table 3: Anti-hormonal therapy: the most common drug groups

Selective estrogen receptor modulators (SERM)	Selective inhibition of estrogen receptors, i.e. a peripheral antiestrogen effect. Indicated in postmenopause with low risk of recurrence or in premenopause with low to moderate risk of recurrence The side-effect profile is partially age-dependent; - of particular relevance: i. Endometrial cancer [(as SERM have a partial estrogen agonist effect on the endometrium; --> G12A) ii. Thromboembolic events iii. Menopausal symptoms
Aromatase inhibitors	Intracellular estrogen synthesis inhibitors. Indication: in postmenopause or combined with GnRH analogues in premenopause Common side effects: i. Increased bone loss ii. Myalgia, arthralgia iii. Menopausal hot flushes
GnRH analogues	Reversible ovarian estrogen production suppression in premenopause. Given in combination with an aromatase inhibitor.
Pure estrogen antagonists	Leads to total estrogen blockade. Only indicated in postmenopause.

Table 4: Anti-hormonal therapy: the most common drug groups (supplements)

Selective estrogen receptor modulators (SERM)	Tamoxifen® 20 mg/d
Aromatase inhibitors	Femara® (letrozole) 2,5 mg/d, Aromasin® (exemestane) 25 mg/d, Arimidex® (anastrozole) 1 mg/d
GnRH analogues	Zoladex® (goserelin), Enantone® (leuprorelin)
Pure estrogen antagonists	e. g. Faslodex® (fulvestrant) 500 mg/d

Adjuvant drug therapy

The choice of treatment is based on HER2 status, hormone receptor status, menopause status and risk profile (risk of recurrence).

If the tumor tissue immunohistochemically expresses hormone receptors, the sequence of chemotherapy and subsequent hormone therapy has an additive effect (the more strongly these hormone receptors are expressed).

Adjuvant anti-hormonal drug therapy

- Only effective with hormone-sensitive tumors (ER and/or PR-positive)!
- Not used - ineffective - for hormone receptor-negative breast cancer.
- Benefit not dependent on patient age or any additional chemotherapy.

Hormone sensitive (ER+) breast cancer with high risk of recurrence. Sequence:

- Adjuvant chemotherapy (see below “Adjuvant chemotherapy”)
- Then adjuvant anti-hormonal drug therapy (type depending on menopause status)
- If necessary, administration of CDK4/6 inhibitor (e.g. abemaciclib, ribociclib) for anti-hormonal therapy

Hormone sensitive (ER+) breast cancer with low risk of recurrence:

- Adjuvant hormone therapy alone (type depending on menopause status)

Adjuvant endocrine therapy for premenopausal women:

- SERM monotherapy for low risk, there is no increased risk of endometrial cancer for menstruating women.
- For high-risk cases and above all in women aged under 35 years or where chemotherapy is indicated, treatment with aromatase inhibitors and GnRH analogues is indicated.

SERM monotherapy for low risk e.g. Tamoxifen® 20 mg/d for 5-10 years

Relevant risk factors regarding the use of aromatase inhibitors:

- N+, large tumor, HER2+, LVI, G3, ER/PR only slightly positive, high Ki-67, invasive lobular cancer, age < 35 or if chemotherapy is indicated

Table 5: Anti-hormonal therapy in early mammaCA [B05] [B06]

	Low risk	Low risk	High risk

	Low risk	Low risk	High risk
Premenopausal	Tamoxifen (5 years)	Tamoxifen or Exemestan + GnRH or Switch (5 years)	Exemestan + GnRH (5-10 years), if necessary, administration of CDK4/6 inhibitor
Postmenopausal	Tamoxifen (5 years)	Up-front AI or Switch (5-10 years)	Up-front AI (5-10 years), if necessary, administration of CDK4/6 inhibitor

Key indications for AI > 5 years: N+ and CTS calculator

Any increase to antihormone therapy must always be discussed with the patient, particularly with regard to side effects and the risk of recurrence.

With aromatase inhibitor treatment, it must also be noted that bone loss is increased by approx. 7 % per year, which can have a dramatic effect on pre-existing osteoporosis.

Several studies (ABCSG12 study, ZO-FAST study, AZURE study, Early Breast Cancer Trialists' Collaborative Group (EBCTCG) see Literature) have shown that the administration of bisphosphonates to women with non-metastatic breast cancer can improve disease-free, recurrence-free and overall survival. As bisphosphonates stabilize/increase bone density, all patients should receive "bone-strengthening" treatment with bisphosphonates e.g. Zometa® (zoledronic acid) 4 mg every six months. In addition, vitamin D3 and calcium therapy should be initiated in all patients.

NOTE: Osteonecrosis as a side effect of bisphosphonates.

- AI treatment: Consider zoledronic acid or oral bisphosphonate for osteoporosis prophylaxis -> establish bone density for all patients
- Premenopausal: consider adjuvant bisphosphonate treatment with hormone ablation (only one study: ABCSG-12!)
- Postmenopausal: recommend adjuvant bisphosphonate treatment (advantage for DFS and OS)

Prerequisite:

- Dental check-up before first administration, as osteonecrosis can be a side effect with pre-existing infection
- Confirmation of health insurance coverage required for bisphosphonates

Adjuvant chemotherapy

With chemotherapy, it is always necessary to find a balance between benefit and toxicity.

Decisions can be guided by prognostic and predictive factors (axillary lymph node involvement, tumor size, Ki-67, grading, age, histological subtype, ER/PR status, vascular invasion, HER2, response to neoadjuvant chemotherapy (pCR)). There are now prognosis tools that can be helpful:

- Adjuvant online
- Nottingham Prognostic index (NPI)
- Genetic signatures such as Oncotype Dx (21 gene score), EndoPredict (70 gene score), MammaPrint (11 gene score), Prosigna (Pam 50)(46 gene score).

With genetic signatures, the tumor is tested for certain genes and a risk of recurrence is calculated (based on studies). [P21]

- Indicated for high risk of recurrence. The benefit here is independent of (any additional) adjuvant hormone therapy, but may be age-dependent: Adjuvant chemotherapy reduces annual breast cancer mortality in women < 50 years old by 40 % and in women aged 50-70 by 20 %.
- Adjuvant chemotherapy is not justified in low-risk cases. This is due to the low level of benefit versus the acute and long-term side effects of chemotherapy (e. g. secondary leukemia, heart failure due to anthracyclines, etc.).
- Implementation according to one of the many standardized treatment regimens.

Hormone sensitive (ER+) breast cancer with high risk of recurrence. Sequence:

- Adjuvant chemotherapy
- Then adjuvant hormone therapy (see above “Drug-based adjuvant hormone therapy”) with administration of CDK4/6 inhibitor, if necessary.

Hormone receptor and HER2-negative breast cancer with high risk of recurrence:

- Adjuvant/neoadjuvant chemotherapy alone

Neoadjuvant chemotherapy

Primary (neoadjuvant) drug treatment is indicated in the following situations:

- Locally advanced breast cancer
- Primary inoperable breast cancer
- Inflammatory breast cancer
- Reduction in size of primary tumor to potentially avoid mastectomy, where the same treatment is indicated postoperative
- As an alternative to postoperative adjuvant treatment indicated due to clinical, imaging and biopsy results.
- The objective is preoperative tumor shrinkage using chemotherapy in order to achieve a better cosmetic result from surgery and to improve response monitoring.
- Patients who do not respond to primary drug-based chemotherapy have a less favorable prognosis [P22, P23].
- The reduction in risk is the same as when chemotherapy is carried out after surgery.
  [P22][P23]
- Neoadjuvant chemotherapy should be considered in particular with cases of HER2+ breast cancer (> 2 cm) and with triple-negative status (ER-, PR-, HER2-), as response is usually good here. Neoadjuvant chemotherapy is combined with immunotherapy (trastuzumab/pertuzumab) in cases of HER2+ breast cancer. In triple negative carcinoma, Pembrolizumab (checkpoint inhibitor) is also added to neoadjuvant chemotherapy according to the KeyNote study522 [P34].
- In cases with triple-negative or HER2+ tumor biology, achievement of pathological complete response (pCR) is associated with improved chances of survival [P22, P23].
  [P22][P23][P34]

Drug-based treatment options in palliative situations

→ see below “Palliative therapy”

Immunotherapy

Adjuvant and neoadjuvant immunotherapy

Until roughly 2000, HER2+ tumors had a poor prognosis. Treatment with trastuzumab was authorized in Europe for patients with metastasized cancer and overexpression of the HER2 growth factor receptor around the year 2000. Monoclonal antibodies have also been in use for patients without metastasis since around 2006. Trastuzumab binds to the epidermal growth factor receptor HER2 (human epidermal growth factor receptor 2) on the cell surface of cancer cells, thus preventing their growth. When administered for one year, this monoclonal antibody-based immunotherapy can significantly lower the rate of recurrence.

There are now different drugs available: Trastuzumab and pertuzumab, which are used alone or in combination depending on tumor size and patient age, combined with chemotherapy. There are also anti-body drug conjugates (ADC): Trastuzumab Emantasin and Trastuzumab Deruxtecan, these are immunoconjugates which consist of a cell-binding component (Trastuzumab) and a cytostatic agent. This ADC can specifically destroy the cell.

- Trastuzumab: Herceptin®
- Pertuzumab: Perjeta®

ADC:

- Trastuzumab emtasin (T-DM1): Kadcyla®
- Trastuzumab Deruxtecan (Enhertu®)

As these drugs have a targeted effect, they also have fewer side effects. The most common side effects are infusion reactions (allergic reactions), impaired liver function, reduced platelet count, neurotoxicity and reversible myocardial damage. Heart function must always be monitoring closely during immunotherapy with monoclonal antibodies.

The NeoSphere and TRYPHAENA studies [P24] showed that significantly higher remission rates could be achieved with the double antibody blockade of trastuzumab (Herceptin®) and pertuzumab (Perjeta®) in neoadjuvant treatment for HER2+ breast cancer (> 2 cm). In the NeoSphere study, adding Perjeta® almost doubled the pCR rate, from 21,5 % to 39,3 %. This is associated with improved chances of survival. [P24]

Palliative immunotherapy

→ see below “Palliative immunotherapy”

Radiotherapy

General notes

- Simultaneous administration of cytostatic agents and radiotherapy (as combined chemoradiotherapy) must be avoided in breast cancer. The sequential use of the two treatments, on the other hand, is acceptable.
- Simultaneous endocrine treatment (SERM or aromatase inhibitor) with radiotherapy is unproblematic and authorized.
- Additive toxicity can lead to pronounced acute and chronic side effects (fibrosis, dermatitis).

Radiotherapy following breast-conserving surgery

Postoperative radiation of the remaining breast is an integral part of breast-conserving treatment. If postoperative radiation is not possible, the indication for breast-conserving surgery should be considered critically.

For breast cancer, the risk of local recurrence after breast-conserving surgery is 15 to 40 % (depending on stage). Radiotherapy reduces this risk to 5 to 10 %. The 20-year survival rate is identical for breast-conserving surgery with radiotherapy of the remaining breast tissue and mastectomy.

The assumption that primary mastectomy with breast reconstruction is advantageous in terms of tumor control or survival versus breast-conserving surgery with subsequent radiotherapy has been disproved.

The conventional radiation dose in the context of breast-conserving treatment is within 25-28 radiation days applied as daily treatment sessions (1). Patients assess the cosmetic results of breast-conserving treatment as good to very good. Hypofractionated radiation can be used to shorten the treatment to approx. 3 weeks (2) with studies showing fewer acute skin reactions in this case. Hypofractionated radiation is the first-line treatment for patients for whom no lymphatic radiation is planned or where there is no elevated risk (3). Boost radiation (saturation of the tumor bed) is additionally recommended for patient with an elevated risk of local recurrence. Boost radiation can be used subsequently to hypofractionated radiation.

- e.g. 1,8 to 2,0 Gy 5x per week, total dose 50 to 50,4 Gy with a boost to the tumor

bed of 5x 2 Gy for 4 to 8 radiation days, thus in total another 10 to 16 Gy
- e.g. 2,66 Gy, 5x per week 15 radiation days, in total 39,9 Gy
- e.g. negative hormone receptors, high grading, etc.

Radiotherapy in high-risk situations following mastectomy

After mastectomy, there is a significant risk (over 50 % depending on stage) of locoregional recurrence in the following situations:

- pronounced axillary lymph node involvement (> 4)
- large tumor diameter (> 5 cm)

The term locoregional recurrence refers to:

- Intramammary recurrence
- Thoracic wall recurrence
- Axillary lymph node/skin recurrence
- Supra/infraclavicular lymph node recurrence

Radiotherapy can reduce the rate of thoracic wall and lymph node recurrence (reduction of expected recurrence by one quarter). Radiotherapy takes place according to the risk profile (N+, > 5cm, R1/R2).

e. g. 2 Gy, 5x per week, 25 radiation days, 50 Gy

The goal of radiotherapy is local tumor control.

Palliative radiotherapy

→ see below “Palliative radiotherapy”

Palliative therapy

In cases with one or more distant metastases, maintaining quality of life (particularly using medication) is the objective, as recovery is not achievable.

Palliative endocrine therapy

Indication

- Metastatic hormone receptor-positive breast cancer
- If possible, the endocrine stage of palliative care should take place sequentially and, if clinically possible, endocrine therapy is preferable to chemotherapy.
- The choice of first-line hormone therapy depends on previous adjuvant hormone therapy (AI/TAM). There are no precise data on the best order, so there is room for variation depending on the clinical situation.
- Consider simultaneous chemotherapy in case of rapid progression or life-threatening organ metastases (e.g. visceral crisis).
- In case of no clinical benefit of 2-3 consecutive endocrine therapies -> chemotherapy.
- In general, all hormone therapy options should be exhausted before chemotherapy is used.

Palliative endocrine therapy for premenopausal women

There are various therapy options:

- GnRH analogues (oophorectomy) + aromatase inhibitor +/-CDK4/6 inhibitor
- If there is a return to tumor progression, the aromatase inhibitor can be replaced by a pure estrogen antagonist* (fulvestrant) with administration of GnRH analogues.

GnRH analogues (oophorectomy) + aromatase inhibitor +/-CDK4/6 inhibitor: e.g. palbociclib (Ibrance©), ribociclib (Kisqali®), abemaciclip (Verzenios®)

Fulvestrant: e. g. Faslodex®

Aromatase inhibitors

- Tamoxifen in combination with hormone ablation treatment (oophorectomy or ovarian suppression with GnRH analogues).
- Tamoxifen alone

The substances palbociclib, ribociclip and abemaciclip are CDK-4/6 inhibitors, which means they block cyclin-dependent kinase 4 and 6 (CDK 4/6), which usually regulate mitosis and cell growth.

Increased CDK-4/6 activity can be found in the tumor tissue of HR+ breast cancer patients, causing the uncontrolled growth of the tumor. The CDK4/6 blockade slows the growth of tumor cells.

Palliative endocrine therapy for postmenopausal women

For postmenopausal women, an aromatase inhibitor is the first-line palliative therapy, or aromatase inhibitor + CDK4/6 inhibitor:

- In case of progression, this can be changed for a combination of exemstane and everolimus or the antiestrogen fulvestrant + CDK4/6 inhibitor.
- Tamoxifen
- Megestat

- CDK4/6 inhibitor: e.g palbociclib (Ibrance©), ribociclib (Kisqali®), abemaciclip (Verzenios®)
- Exemstane: Aromasin®
- Everolimus: 10 mg /d, if necessary phased increase
- Antiestrogen fulvestrant: Faslodex® 500 mg/d

Note: SERM can also be used as palliative therapy for postmenopausal women, although several studies demonstrate the superiority of aromatase inhibitors.

Palliative chemotherapy

Indications:

- Rapid tumor progression
- Extensive metastasis in parenchymatous organs (“visceral crisis” - e. g. liver metastases, lung metastases, bone marrow cancer)
- Metastasized hormone receptor-negative breast cancer

The palliative chemotherapy regimen selected is adapted to the symptoms:

- life-threatening symptoms --> polychemotherapy (taking into account quality of life)
- mild symptoms --> monochemotherapy

Palliative immunotherapy

For breast cancer with HER2 overexpression, immunotherapy can be given with the monoclonal antibody trastuzumab in combination with pertuzumab and cytostatics (e.g. taxanes) [P24, P25].

- The response rate to immunotherapy alone is much lower.
- Trastuzumab in combination with antiestrogen therapy is much less effective and remains reserved for individual cases.
- The new substance TDM1 or trastuzumab deruxtecan can be given as subsequent treatment with good response and tolerability.
- A cytostatic with a tyrosine kinase inhibitor can also be given as a further line of treatment e.g. lapatinib/capecitabine
- A combination of cytostatics with trastuzumab can also be given subsequently. [P24][P25][P29]

- Trastuzumab: Herceptin®
- Pertuzumab: Perjeta®
- Substance TDM1: Kadcyla® 3,6 mg/kg, every 3 weeks
- Trastuzumab Deruxtecan (Enhertu®) by Lapatinib: Tyverb®
- Capecitabine: Xeloda®

Immunotherapy is an important treatment option for palliative care in HER2+ metastasized breast cancer.

Immunotherapy is an important treatment option for palliative care in HER2+ metastasized breast cancer.

Govitecan sacituzumab binds specifically to the Trop2 glycoprotein, a special receptor (Trop-2 receptor) on the surface of the tumor cell. After binding to the receptor, the immunoconjugate is internalized. The linker SN-38, which interacts with topoisomerase I, is then hydrolyzed. This interaction prevents the reclosure of single-strand breaks induced by topoisomerase I. The resulting DNA damage leads to cell death [P33]. [P33]

Palliative radiotherapy

Local recurrences can be tackled curatively. The objective is total tumor control, as well as localized pain relief. Palliative care is of particular value for metastasis affecting bone only, as the course of the disease can be very long in the absence of visceral metastasis. For bone-related symptoms, short-term radiation can provide pain relief in 75 % of cases. Other important palliative indications for radiotherapy include brain metastases (carried out under corticosteroids to prevent brain edema!).

short-term radiation: 5 x 4 Gy or 10 x 3 Gy

Follow-up

After treatment, follow-up is necessary!

Objective of follow-up:

1. Identification of local recurrence
2. Identification of any secondary cancer on opposite side
3. Identification of side effects from adjuvant treatment that can be alleviated
4. Clarification of symptoms that could suggest distant metastasis
5. Psychosocial care.

In the first and second year, medical follow-up takes place every 3 to 6 months, then roughly every 6 months until the fifth year, then annually.

Follow-up consultation

- Ask about and if necessary clarify any symptoms.
- Clinical examination: Precise locoregional status examination (standing and lying) including opposite side.
- Annual mammography (if necessary ultrasound), consider annual MRI for dense breast tissue and younger patients.

If no symptoms: no laboratory tests, no CA 15-3, no chest x-ray, no abdominal ultrasound, no bone scan.

Annex

Notes on treatment of metastasis

- Solitary brain metastasis or epidural metastasis: surgery if possible, then radiation.
- Multiple brain metastases: Immediate treatment with steroids to reduce perifocal edema. Then whole brain radiation therapy.
- Painful bone metastases: Sufficient drug-based pain therapy. Radiotherapy is very efficient for pain relief.
- Weight bearing skeletal areas at risk of fracture may require osteosynthetic stabilization. For osteolytic bone metastases, bisphosphonates or specific antibodies are used (monthly infusion) to limit the progression of skeletal deterioration. Bisphosphonates can thus prevent pathological fractures and contribute to pain relief. They are also effective against paraneoplastic hypercalcemia.

Further information on surgical techniques for breast reconstruction

From an aesthetic perspective, the surgeon does the following:

- Compensation of skin deficit.
- Restoration of breast contour.
- Reconstruction of nipple-areola complex.
- If necessary, achievement of symmetry through alignment of the opposite side (after 4 to 6 months).

Surgery to adjust the opposite side is not always carried out at the same time as breast reconstruction, as the reconstructed breast can change shape and position during healing and after radiotherapy. This also applies to the reconstruction of the nipple-areola complex.

Since December 2014, alignment of the opposite side has been covered by basic health insurance for breast cancer patients in Switzerland.

Despite all the techniques currently available, it is not yet possible to reconstruct the nipple-areola complex with all its structures. Some techniques use full skin grafts from pigmented areas of skin (such as the labia). Reconstructing a 3D nipple is very challenging. A nipple can also be restored using tattooing.

A variety of reconstruction techniques are used by surgeons. The most common are:

- Implantation of silicone prostheses: This requires sufficient well-perfused skin covering. The implant is inserted subpectorally or above the pectoralis muscle and if necessary covered with a mesh. Disadvantage of silicone prostheses: Increased capsular contracture after radiation.
- Implantation of an expander silicone prosthesis: If the skin is tight or radiotherapy is planned, the skin can first be expanded, in proportion to the healthy breast, in order to later cover the prosthesis. The expander prosthesis is generally replaced by a definitive silicone prosthesis in a second surgery (after 2 to 3 months).
- Flap techniques: Previous scarring and physical build influence the surgical technique.

Note: There are differences in terms of indication and technique between skin flap, musculofascial flap and musculocutaneous flap techniques, particularly with regard to additional relining with a silicone prosthesis.

Additional specific procedures include:

- Free flap techniques such as DIEP (Deep Inferior Epigastric artery Perforator flap), TMG (Transverser Muskulus Gracilis flap) flap techniques etc. depending on the physique. It is even possible to use a vascularized combined tissue block (as described above). This can then be positioned freely in the breast area - followed by microvascular anastomosis to the internal thoracic artery and internal thoracic vein, for example, under microscope.
- TRAM flap (Transverse Rectus Abdominis Muscle Flap): A tissue block of skin, fat and rectus abdominis muscle is made from the abdomen (at the superior epigastric artery) and then pivoted in a cranial direction either homolaterally or contralaterally.
- Pedicle latissimus dorsi flap (a musculocutaneous flap).

Further information on imaging diagnostics for breast cancer

Mammography

Mammography has specific technical requirements in terms of radiation quality, positioning technique and imaging system. This is due to the structure of the female breast, which comprises exclusively soft tissue; - the difference in contrast between the different tissues is therefore low. It must be possible to depict and identify microcalcifications measuring 0,1 mm. As the breast is round, it must be compressed. Patients often find this unpleasant. This compression is important for positioning the breast and avoiding motion artifacts.

It changes the breast shape to a cube, so that contrasts are balanced out. On mammography, cancer can appear as a group of microcalcifications or an opacification, typically as a star with “spikes” extending out.

Breast ultrasound

Breast ultrasound is carried out with a linear probe at a frequency of 11 to 13 mHz. Breast cancer is usually a hypoechoic lesion with indistinct margins with distorted Cooper’s ligaments and some acoustic attenuation.

Further information on stereotactic biopsy methods

In case of non-palpable suspicious changes on mammography not depicted on ultrasound (suspicious groups of microcalcification, suspicious opacity), stereotactic biopsy is the first-line method!

For this, the patient lies prone on a special x-ray table with the affected breast hanging through an opening, so that the breast is accessible from all sides. The breast is compressed on a digital x-ray plate and a mammography is taken from various angles with a pivotable x-ray tube. The computer can use the different angles to establish the depth and precise location of the lesion (stereotaxy).

The biopsy needle, which is connected to the computer, then received the precise coordinates of the lesion and is inserted into the breast at the right location under local anesthetic. Tissue is sucked into the hollow needle using vacuum aspiration and then detached using a core needle (tissue cylinder) for histological testing. Lesions measuring up to 1 cm can be removed.

If a breast MRI gives an abnormal finding that is not visible on US or mammography, an MR-controlled vacuum biopsy is carried out. In order to locate the lesion again, a contrast is injected into a vein in the arm as in the previous MRI. The precise position of the lesion is then calculated and the location biopsied with vacuum aspiration.

Further information on treatment in case of BRCA1 or BCA2 mutation

All patients with high-risk family history (> 30 % lifetime risk) or triple-negative breast cancer < 40 years old should undergo genetic testing for BRCA1 and BRCA2 mutations after confirmation of insurance coverage.
For BRCA+, metastasized or locally advanced, HER2- breast cancer, treatment with olaparib tablets has been authorized as pre-treatment since July 2019. Olaparip inhibits the Poly-ADP-Ribose-Polymerase (PARP) enzyme and thus prevents cancer cells from repairing DNA damage cause by cytostatics. PARP inhibitors are therefore primarily used after chemotherapy maintenance treatment. The PARP inhibitor olaparib is also used adjuvantly after (neo)adjuvant chemotherapy in primary BRCA1/2-associated HER2-negative breast cancer according to the OlympiA study [B07].
Based on several further studies, the approval for treatment with olaparib will certainly be applied to other indications in the future (see also ovarian cancer). [B07]

Treatment with olaparib (Lynparza®) tablets: 150 mg 2-0-2

Staging

Note: deliberately brief and simplified!

Table 6: Primary tumor [B07] [B07]

TX	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis (DCIS)	Ductal carcinoma in situ
Tis (Paget)	Paget disease of the nipple (epidermal adenocarcinoma in situ), Paget disease associated with a tumor is classified according to the size of the tumor!
T1	Microinvasion (substage T1mic: max. 1 mm) or invasion > 1 mm to max. 20 mm (Substages T1a-c).
T2	Tumor size 20,1–50 mm
T3	Tumor size > 50 mm
T4a	Extension to chest wall (not including pectoralis muscle)
T4b	Edema or ulceration of the skin or satellite skin metastases of the same breast
T4c	Both T4a and T4b
T4d	Inflammatory breast cancer

Table 7: Regional lymph nodes according to histological findings

pN0	No axillary lymph nodes metastasis histologically
pN1mi	Micrometastasis 0,2–2 mm
pN1a	Metastasis in 1–3 axillary lymph nodes
pN1b	Microscopic metastasis in internal mammary lymph nodes detected by sentinel lymph node biopsy but not clinically apparent
pN1c	Both pN1a and pN1b

pN2a	Metastasis in 4–9 axillary lymph nodes (at least one > 2 mm)
pN2b	Clinically apparent internal mammary lymph node metastasis in the absence of axillary lymph node metastasis
pN3a	Metastasis in 10 or more axillary lymph nodes (at least one > 2 mm) or infraclavicular lymph node metastasis
pN3b	Clinically apparent ipsilateral internal mammary lymph node metastasis and axillary lymph node involvement; or more than 3 positive axillary lymph nodes plus microscopic internal mammary metastasis detected by sentinel lymph node biopsy
pN3c	Metastasis in ipsilateral supraclavicular lymph nodes

Table 8: Distant metastasis

MX	Presence of distant metastasis cannot be assessed
M0	No distant metastasis
M1	Distant metastasis

Literature

Nr.	Title	Author	Journal	Volume	P
[B01]	Management des Mammakarzinoms	Kreienberg, Jonat, Volm, Möbus, Alt	Onkologie aktuell, Springer Verlag	3. Auflage	
[B02]	Mammakarzinom: Interdisziplinär	Kreienberg, Möbus, Jonat, Kühn	Springer Verlag	4. Auflage	
[B03]	Aktuelle Empfehlungen zur Prävention, Diagnostik und Therapie primärer und fort-geschrittener Mammakarzinome	Kommission Mamma der AGO e.V.	State of the Art 2011	5. Auflage	3
[B04]	Interdisziplinäre S3-Leitliniefür die Früherkennung, Diagnostik, Therapie und Nachsorge des Mammakarzinoms	n.a.		AWMF-Registernummer: 032-045OL	
[B05]	Adjuvant Exemestane with Ovarian Suppression in Premenopausal Breast Cancer	Pagani et al.	N Engl J Med	371	1
[B06]	Standard Operating Procedure (SOP)	Universitätsspital Basel			

Nr.	Title	Author	Journal	Volume	P
[B07]	Olaparib for Metastatic Breast Cancer in Patients with a Germline BRCA Mutation	Robson M, Im S, Senkus E, Xu B, Domchek SM, Masuda N, Delaloge S, Conte P	N Engl J Med	377	5

Nr.	Title	Author	Journal	Volume	Pages
[P01]	Twenty-year follow-up of a randomized trial comparing total mastectomy, lumpectomy, and lumpectomy plus irradiation for the treatment of invasive breast cancer	Fisher B, Anderson S, Bryant J, Margolese RG, Deutsch M, Fisher ER, Jeong JH, Wolmark N	N Engl J Med	17;347(16)	1233-1241
[P02]	Impact of Radiotherapy on Complications and Patient-Reported Outcomes After Breast Reconstruction	Jagsi et al.	J Nati Cancer Inst	110	

Nr.	Title	Author	Journal	Volume	Pages
[P03]	Axillary Dissection vs No Axillary Dissection in Women With Invasive Breast Cancer and Sentinel Node Metastasis. A Randomized Clinical Trial	Giuliano AE, Hunt KK, Ballman KV, Beitsch PD, Whitworth PW, Blumencranz PW, Leitch AM, Saha S, McCall LM, Morrow M	JAMA	Vol 305, No. 6	
[P04]	Sentinel lymph node biopsy in breast cancer: A history and current clinical recommendations	D'Angelo-Donovan DD, Dickson-Witmer D, Petrelli NJ	Surg Oncol		
[P05]	Axillary lymph node dissection in early-stage invasive breast cancer: is it still standard today?	Gerber B, Heintze K, Stubert J, Dieterich M, Hartmann S, Stachs A, Reimer T	Breast Cancer Res Treat	128(3)	613-624

Nr.	Title	Author	Journal	Volume	Pages
[P06]	Axillary dissection versus no axillary dissection in patients with breast cancer and sentinel-node-micrometastases (IBCSG 23-01): 10-year follow-up of a randomized, controlled phase 3 trial	Galimberti V et al.	Lancet Oncol		
[P07]	Effect of axillary dissection vs no axillary dissection on 10-year overall survival among women with invasive breast cancer and sentinel node metastasis: the ACOSOG Z0011 (Alliance) randomized clinical trial	Giuliano et al.	JAMA	318	918-926

Nr.	Title	Author	Journal	Volume	Pages
[P08]	Interpreting Breast International Group (BIG) 1-98: a randomized, double-blind, phase III trial comparing letrozole and tamoxifen as adjuvant endocrine therapy for postmenopausal women with hormone receptor-positive, early breast cancer	Regan MM, Price KN, Giobbie-Hurder A, Thürlimann B, Gelber RD; International Breast Cancer Study Group and BIG 1-98 Collaborative Group	Breast Cancer Res	13(3)	209
[P09]	Assessment of letrozole and tamoxifen alone and in sequence for postmenopausal women with steroid hormone receptor-positive breast cancer: the BIG 1-98 randomised clinical trial at 8•1 years median follow-up	Regan MM, Neven P, Giobbie-Hurder A, Goldhirsch A, Ejlertsen B, Mauriac L, Forbes JF, Smith I, Láng I, Wardley A, Rabaglio M, Price KN, Gelber RD, Coates AS, Thürlimann B; BIG 1-98 Collaborative Group; International Breast Cancer Study Group (IBCSG)	Lancet Oncol	12(12)	1101-1108

Nr.	Title	Author	Journal	Volume	Pages
[P10]	Longer-Term Outcomes of Letrozole Versus Placebo After 5 Years of Tamoxifen in the NCIC CTG MA.17 Trial: Analyses Adjusting for Treatment Crossover	Jin H, Tu D, Shepherd LE, Goss PE	J Clin Oncol		
[P11]	Effect of anastrozole and tamoxifen as adjuvant treatment for early-stage breast cancer: 10-year analysis of the ATAC trial	Cuzick J, Sestak I, Baum M, Buzdar A, Howell A, Dowsett M, Forbes JF; ATAC/LATTE investigators	Lancet Oncol	11(12)	1135-1141

Nr.	Title	Author	Journal	Volume	Pages
[P12]	Which patients benefit most from adjuvant aromatase inhibitors? Results using a composite measure of prognostic risk in the BIG 1-98 randomized trial	Viale G, Regan MM, Dell'Orto P, Mastropasqua MG, Maiorano E, Rasmussen BB, MacGrogan G, Forbes JF, Paridaens RJ, Colleoni M, Láng I, Thürlimann B, Mouridsen H, Mauriac L, Gelber RD, Price KN, Goldhirsch A, Gusterson BA, Coates AS; BIG 1-98 Collaborative and International Breast Cancer Study Groups	Ann Oncol	22(10)	2201-2217
[P13]	Influence of comorbidities and age on risk of death without recurrence: a retrospective analysis of the Arimidex, Tamoxifen Alone or in Combination trial	Ring A, Sestak I, Baum M, Howell A, Buzdar A, Dowsett M, Forbes JF, Cuzick J	J Clin Oncol	29(32)	4266-4272

Nr.	Title	Author	Journal	Volume	Pages
[P14]	The advantage of letrozole over tamoxifen in the BIG 1-98 trial is consistent in younger postmenopausal women and in those with chemotherapy-induced menopause	Chirgwin J, Sun Z, Smith I, Price KN, Thürlimann B, Ejlertsen B, Bonnefoi H, Regan MM, Goldhirsch A, Coates AS; for the BIG 1-98 Collaborative and International Breast Cancer Study Groups	Breast Cancer Res Treat	131(1)	295-306
[P15]	Adjuvant Exemestan with Ovarian suppression in premenopausal Breast Cancer	Pagani O et al.	N Engl Med	371	107-118
[P16]	Adjuvant bisphohonate treatment in early breast cancer: meta-analyses of individual patient data from randomized trials	EBCTCG	Lancet	3386	1353-1361

Nr.	Title	Author	Journal	Volume	Pages
[P17]	Effectiive Inhibition of aromatase inhibitor-associated bone loss by zoledronic acid in postmenopausal women with early breast cancer receiving adjuvant letrozole: ZO-FAST Study results	Bundred NJ et al.	Cancer	115	1001-1010
[P18]	Zoledronic acid combined with adjuvant endocrine therapy of tamoxifen versus anastrozol plus ovarian suppression in premenopausal early breast cancer: final analysis of Austrian Breast and Colorectal Cancer Study Group Trial 12	Gnant et al.	Ann Oncol	26	313-320

Nr.	Title	Author	Journal	Volume	Pages
[P19]	Benefits and risks of adjuvant treatment with zoledronic acis in stage II/III breast cancer. 110 years follow-up of the AZURE randomized clinical trial (BIG 01/04)	Coleman RE et al.	J Bone Oncol	13	123-135
[P20]	A gene expresiion signature as a predictor of survival in breast cancer	Van De Vijivern MJ et al.	N Engl J Med	347	1999-2009
[P21]	A Multigene assay to predict recurrence of tamoxifen treated-, node-negative breast cancer	Paik et al.	N Engl. J Med	351	2817-2826

Nr.	Title	Author	Journal	Volume	Pages
[P22]	Use of Biomarkers to Guide Decisions on Adjuvant Systemic Therapy for Women with Early-Stage Invasive Breast Cancer: ASCO clinical Practice Guideline Update-Integration of Results from TAILORx	Andre et al.	J Clin Oncol	37	1965-1964
[P23]	Pathological complete response and long-term clinical benefit in breast cancer: the CTNeoBC pooled analysis	Cortazar P, Zhang L, Untch M et al.	Lancet	384(9938)	164–172
[P24]	Pathological complete response in neoadjuvant treatment of breast cancer	Cortazar P, Geyer CE	Ann Surg Oncol	22	1441-1446

Nr.	Title	Author	Journal	Volume	Pages
[P25]	Efficay and safety of neoadjuvant pertuzumab und trastuzumab in women with locally advanced, inflammatory, or early HER2 positive breast cancer (Neo Sphere): a randomised multicentre, openlabel, phase 2 tiral	Gianni L et al.	Lancet Oncol	13	25-32
[P26]	Pertuzumab plus trastuzumab in combination with standard neoadjuvant antracycline-containing and antrhacycline-free chemotherapy regimens in patients with HER”-positive early breast cancer: a randomized phase II cardiac safety study (TRYPHAENA).	Schneeweiss et al.	Ann Oncol	24	227-284

Nr.	Title	Author	Journal	Volume	Pages
[P27]	Tratuzumab Emantasine for Residual Invasive HER2-Positive Breast Cancer	Von Minckwitz G et al.	N Engl J Med	380	617-628
[P28]	The cyclin-dependent kinase 4/6 inhibitor palbociclib in cominbation with letrozole versus letrozole alone as first-line treatment of oestrogen recetor-positive, HER2-negative, advanced breast cancer (PALOMA-A/TRIO): a randomized phase 2 study	Finn RS et al.	Lancet Oncol	16	25-35
[P29]	Pertuzumab, trastuzumab, and docetaxel in HER2-positive metastatic breast cancer	Swain SM et al.	N Engl J. Med	372	324-334

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[P30]	Trastuzumab emantsine for HER2 advanced breast cancer	Verma S et al.	N Engl J Med	367	1783-1791
[P31]	Olabarib for Metastatic Breast Cancer in Patients with Germline BRCA Mutation	Robson M et al.	N Engl J Med	377	523-533
[P32]	Adjuvant Olaparib for Patients with BRCA1-or BRCA2 Mutated Breast Cancer	Tutt ANJ et al.	N Engl J Med	384(25)	2394-2405
[P33]	Sacituzumab Govitecan in Metastatic Triple-Negative Breast Cancer	Bardia A et al.	N Engl J Med	384(16)	1529-1541
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Nr.	Title	Author	Journal	Volume	Pages
[P35]	Ribociclib and endocrine therapy as adjuvant treatment in patients with HR+/HER2- early breast cancer: Primary results from the phase III NATALEE trial	Slamon DJ, Stroyakovskiy D, Yardley DA, Huang C, Fasching PA, Crown J, Bardia A, Gabriel N et al.	Journal of Clinical Oncology	41	